

BALLATER AGCM, AB, Canada

WMO: 715415

Lat: 55.5769N

Lon: 117.2356W

Elev: 1837

StdP: 13.75

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.8	-23.2	-35.1	0.8	-28.5	-28.9	1.2	-23.0	24.9	34.4	22.2	25.4	4.5	150	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	82.7	62.7	79.4	61.4	76.4	60.2	66.2	77.5	64.2	75.4	62.4	73.0	8.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	88.6	71.5	59.9	82.6	68.9	58.2	77.3	66.6	31.9	77.7	30.3	75.8	29.0	73.3	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	19.0	16.8	-37.4	88.0	3.5	2.3	-39.9	89.7	-42.0	91.1	-44.0	92.4	-46.6	94.1		
(5)			-37.6	70.0	3.5	2.6	-40.1	71.8	-42.2	73.3	-44.2	74.7	-46.8	76.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	12.0	10.2	21.5	37.2	52.6	58.9	61.9	60.2	51.3	38.5	19.7	9.9
	DBStd	22.86	16.85	17.11	15.28	9.79	7.63	5.76	4.54	5.46	8.14	8.57	13.96	14.66
	HDD50	6233	1178	1114	884	391	62	6	0	2	80	362	910	1244
	HDD65	10531	1643	1534	1349	834	387	197	118	166	414	820	1360	1709
	CDD50	1232	0	0	0	7	142	271	368	318	119	7	0	0
	CDD65	55	0	0	0	0	2	13	21	16	3	0	0	0
	CDH74	1217	0	0	0	7	143	244	391	317	115	0	0	0
	CDH80	224	0	0	0	2	28	39	75	61	19	0	0	0
	WSAvg	7.7	7.8	7.2	7.4	9.1	9.1	8.6	6.5	6.5	7.2	7.9	7.6	7.4
(15) Precipitation	PrecAvg	18.7	1.3	0.9	0.7	0.7	1.7	3.0	2.8	2.4	1.5	1.1	1.2	1.2
	PrecMax	27.2	3.5	1.8	1.5	1.4	4.3	5.9	4.4	5.1	4.6	2.5	2.6	2.6
	PrecMin	11.3	0.4	0.3	0.1	0.1	0.5	1.3	0.4	0.7	0.1	0.3	0.3	0.2
	PrecStd	3.8	0.9	0.4	0.4	0.4	1.1	1.3	1.3	1.1	1.0	0.5	0.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	45.0	53.2	70.6	83.7	84.7	86.4	85.6	82.4	67.4	53.0	43.3
		MCWB	39.0	38.8	44.5	52.8	56.7	62.2	66.5	65.4	61.4	51.1	43.9	36.8
	2%	DB	42.6	41.3	47.8	62.9	77.9	79.9	81.7	81.3	77.2	60.2	46.1	39.5
		MCWB	37.4	36.1	40.7	46.3	54.6	60.3	64.8	64.3	58.7	48.4	39.9	34.4
	5%	DB	40.1	38.1	43.3	57.5	73.4	76.5	78.2	77.5	72.0	56.2	42.1	36.4
		MCWB	35.6	34.4	37.7	44.3	53.3	59.0	63.3	62.4	56.6	46.0	37.1	32.1
	10%	DB	37.0	34.2	39.9	52.6	69.8	73.1	75.3	74.1	66.6	52.0	38.0	32.3
		MCWB	33.7	31.6	35.3	41.8	51.2	57.8	61.6	60.9	53.9	43.9	34.5	29.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	39.3	46.2	52.6	58.7	65.5	70.4	68.2	63.6	53.6	44.6	37.5
		MCDB	44.5	44.2	52.3	69.3	74.4	78.5	82.2	79.7	80.6	64.9	51.2	42.3
	2%	WB	37.6	36.7	41.1	47.4	56.2	62.8	67.0	65.7	59.4	49.5	40.4	34.9
		MCDB	42.3	40.6	47.2	60.6	72.7	75.5	77.7	77.8	73.6	58.0	45.5	39.1
	5%	WB	35.9	34.6	37.9	44.6	54.4	60.7	64.9	63.8	57.3	47.1	37.3	32.3
		MCDB	39.8	37.9	42.9	55.9	70.5	73.4	75.2	75.0	70.0	54.2	41.7	36.2
	10%	WB	33.8	31.6	35.4	42.3	52.6	58.9	63.0	62.1	55.1	44.7	34.6	29.0
		MCDB	36.8	34.5	39.6	51.8	67.5	69.6	72.8	72.3	65.2	51.0	37.9	32.3
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	20.4	19.9	20.6	26.0	21.3	22.4	24.2	23.8	18.8	17.1	17.5
		MCDBR	18.7	20.6	20.0	29.5	33.6	29.2	29.8	29.4	33.0	26.2	20.0	21.6
		MCWBR	14.5	16.0	13.8	16.3	15.5	13.1	15.5	14.7	16.1	15.6	14.3	17.8
		MCDBR	18.2	19.8	19.7	27.2	30.8	25.4	27.2	27.5	30.8	24.0	19.3	21.0
	5% WB	MCWBR	14.4	15.7	13.7	15.5	14.7	12.3	15.1	14.9	15.6	14.9	14.2	17.5
(40) Clear-Sky Solar Irradiance	taub		0.240	0.249	0.289	0.336	0.347	0.353	0.354	0.353	0.312	0.286	0.248	0.236
	taud		2.308	2.365	2.349	2.354	2.390	2.396	2.418	2.400	2.523	2.558	2.431	2.220
	Ebn at Noon		227	271	282	281	284	283	280	272	272	256	226	194
	Edh at Noon		16	22	29	34	35	35	34	32	25	19	14	14
(44) All-Sky Solar Radiation	RadAvg		242	561	1056	1498	1776	1919	1877	1531	1037	561	255	169
	RadStd		23	45	85	95	129	145	122	123	87	67	19	15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A

Nomenclature: See separate page